
MELANOMA CLASSIFICATION WITH LIGHTWEIGHT MODIFIED CNNs: A STUDY ON ISIC SLICE-3D AND PATIENT-LEVEL UGLY DUCKLING ANALYSIS

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ABSTRACT

Melanoma is a highly lethal form of skin cancer, making early detection essential. In this project, we investigate whether a lightweight convolutional neural network (CNN) trained from scratch can effectively classify melanoma in the highly imbalanced SLICE-3D dataset, which contains approximately 400.000 lesion crops with fewer than 1% malignant cases. We compare a simple baseline CNN with a modified architecture incorporating batch normalization and global average pooling, train both using class-weighted loss, and evaluate the effect of malignant oversampling. To extend beyond individual lesions, we additionally explore two patient-level “ugly duckling” strategies based on feature-space outlier detection and predicted malignancy probabilities. Across all experiments, the models struggled to detect malignant lesions, with the best model achieving an AUC of 0.6663 but identifying no true positives at clinically relevant operating points. Oversampling provided no meaningful benefit, and patient-level results reflected the limitations of the underlying lesion-level classifier. These findings suggest that small CNNs trained from scratch are insufficient for melanoma detection in SLICE-3D, and that deeper or pretrained architectures are likely required for reliable performance.

Keywords melanoma · skin lesion classification · convolutional neural network · deep learning · SLICE-3D dataset · data augmentation · class imbalance · ugly duckling · patient-level analysis · lightweight CNN

1 Introduction

Melanoma is among the most dangerous forms of skin cancer, and early detection is strongly linked to improved clinical outcomes. Automated melanoma detection has been an active research area for more than a decade, supported by publicly available datasets through the ISIC challenges [Codella et al. 2019, Gutman et al. 2018]. These competitions have historically demonstrated that deep architectures (e.g. ResNet-based systems, ensembles, and models pretrained on large image corpora) achieve the strongest performance.

However, the newly introduced SLICE-3D dataset [Tschandl et al. 2024], which contains hundreds of thousands of lesion crops extracted from 3D total-body photography, presents new challenges: severe class imbalance, wide variation in crop quality, and limited per-patient malignant cases. While high-capacity neural networks can leverage such data, training them requires substantial GPU memory, which may not always be available.

In this project, we explore whether a lightweight modified CNN can learn discriminative features for melanoma classification. We additionally evaluate patient-level predictions using the “ugly duckling” hypothesis, in which suspicious lesions deviate from the patient’s baseline appearance, such as birthmarks.

2 Related Work

The ISIC melanoma detection challenges [Gutman et al., 2018, Codella et al., 2019] have shown consistent progress in automated skin lesion analysis. Top performing approaches often rely on deep CNN architectures, pretrained feature extractors, or ensemble strategies. These models benefit from large-scale training data and computational resources.

With the introduction of SLICE-3D [Tschandl et al., 2024], a large-scale dataset of lesion crops obtained from 3D total-body photography was provided. Its purpose is to support research on broader patient-level screening, not just a diagnosis based on a single image. The companion challenge website [isi a] and Kaggle platform [isi b] describe the classification tasks and highlight the extreme malignant-to-benign ratio present in the dataset.

To our knowledge, no published work has yet to demonstrate strong performance using small, lightweight CNNs on SLICE-3D without the use of transfer learning. This project therefore serves as an exploration of feasibility under constrained resources.

3 Methods

3.1 The Dataset

We use the full SLICE-3D dataset [Tschandl et al., 2024], which provides metadata, the training images, and their classes. The full dataset contains approximately 400,000 lesion crops, of which only about 400 are **malignant** and the rest **benign**, making the dataset extremely imbalanced, with fewer than 1% of lesions being malignant. Examples of lesion crops can be seen in Figure 1

Each image includes:

- A lesion crop from a patient-level 3D skin scan
- A corresponding binary label (benign vs. malignant)
- A patient ID enabling multi-lesion grouping

The images also contain multiple other features, but the majority of those were not used during this research.

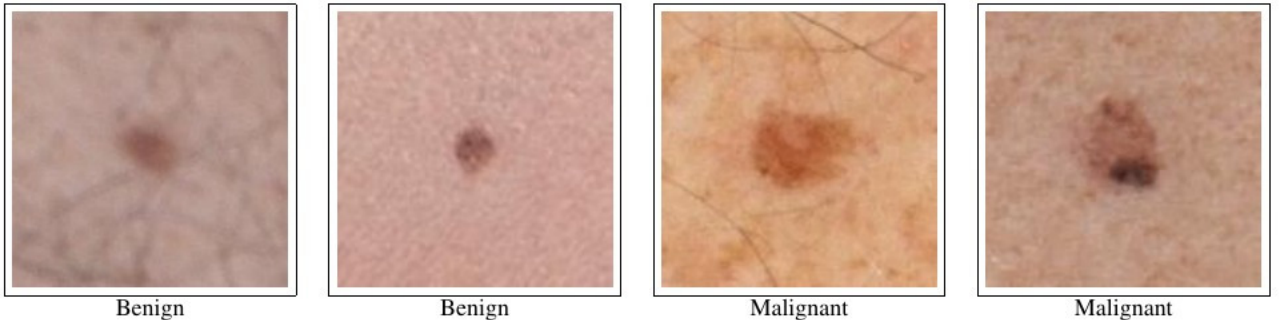


Figure 1: **Example lesion crops from SLICE-3D.** The images illustrate the visual variability in lesion appearance, skin tone, and background texture observed in the dataset.

The biggest factor, as well as difficulty, of this dataset is the **extreme class imbalance**, with malignant lesions representing a very small percentage of the dataset. We do not use undersampling, as early tests showed that discarding benign samples reduced generalization in combination with methods explained further within the report, such as class-weighted loss.

3.2 Preprocessing

All lesion crops were loaded as RGB images and resized to 224×224 pixels to match the input requirements of our convolutional networks. Pixel values were normalized using standard ImageNet statistics, a preprocessing step that has consistently been shown to improve convergence when training CNNs on natural images.

During training, we applied a series of geometric and photometric augmentations to increase robustness and reduce overfitting. The SLICE-3D dataset exhibits substantial variability in illumination, camera angle, and crop tightness due

to the nature of 3D total-body photography. To model these sources of variation, we introduced random horizontal and vertical flips as well as small rotations, which help the network learn orientation-invariant features. We also included random crops and mild perspective changes to mimic the distortions that arise when projecting curved body surfaces onto 2D images.

Color based augmentations, such as brightness and contrast jitter, were used to account for differences in lighting conditions and skin tone across patients. We also incorporated Gaussian blur and random erasing, which mimic common artifacts such as motion blur, slight defocus, or small occlusions. Together, these augmentations nudge the model toward relying on broader structural patterns rather than fine-grained pixel details. Examples of these augmentations can be seen in Figure 2

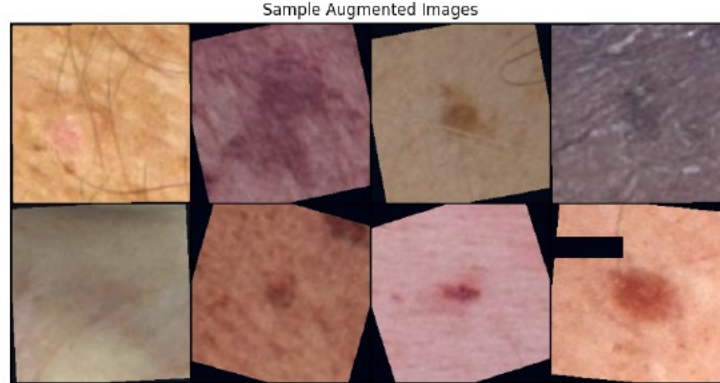


Figure 2: **Augmented lesion crop samples.** Samples from the dataset after performing some augmentations

Because malignant lesions make up less than one percent of the dataset, they appear repeatedly across epochs even without explicit oversampling. In combination with augmentation, this repeated exposure effectively increases their visual diversity during training. This helps the model avoid simply memorizing the minority samples, while still preventing the severe distribution shift introduced by heavy synthetic oversampling.

The dataset was then split into training, validation, and test sets, with 60% of the images used for training, 20% for validation, and 20% for testing. This split ensures sufficient samples for learning while retaining independent sets for hyperparameter tuning and evaluation.

3.3 Baseline CNN

Our first model serves as a simple baseline to verify that the end-to-end training pipeline – data loading, augmentation, loss weighting, batching, and evaluation – functions correctly. The architecture, shown in Table 1, consists of three convolutional blocks followed by a small fully connected classifier. Each block contains a 3×3 convolution, ReLU activation, and 2×2 max-pooling, progressively reducing the spatial resolution.

The model contains approximately **11 million parameters**. While small compared to modern CNNs, our Baseline CNN is still underpowered for the SLICE-3D dataset, which features highly variable crops, subtle pigmentation features, and extreme class imbalance. Because its convolutional blocks use only ReLU activations and max-pooling, the model has limited capacity to capture melanoma-specific features such as pigment networks or irregular borders.

3.4 Modified CNN

To strengthen the baseline architecture while maintaining low computational cost, we designed a modified CNN inspired by the structure of lightweight dermatology models. The key changes relative to the baseline are:

- **Batch Normalization** after each convolution, improving optimization stability [Ioffe and Szegedy 2015].
- **Global Average Pooling (GAP)** to reduce the number of fully connected parameters and encourage more spatially robust feature learning.

Table 1: Baseline CNN architecture.

Layer	Output shape
Input	$3 \times 224 \times 224$
Conv(3,32) + ReLU	$32 \times 224 \times 224$
MaxPool(2)	$32 \times 112 \times 112$
Conv(32,64) + ReLU	$64 \times 112 \times 112$
MaxPool(2)	$64 \times 56 \times 56$
Conv(64,128) + ReLU	$128 \times 56 \times 56$
MaxPool(2)	$128 \times 28 \times 28$
Flatten	100.352 features
FC(100352 $\rightarrow$ 256) + ReLU + Dropout(0.5)	256
FC(256 $\rightarrow$ 2)	logits

- A significantly smaller classifier head ($128 \rightarrow 256 \rightarrow 2$ instead of tens of thousands of units).

The overall architecture is shown in Table 2. While the convolutional stack has a similar depth to the baseline, BatchNorm layers have been added after each convolution to stabilize feature distributions and speed up training. After the final convolutional stage, a global average pooling (GAP) layer produces a compact 128-dimensional feature vector for each image. This replaces the large flattening operation used in the baseline model, significantly reducing the number of parameters and helping to mitigate overfitting.

Table 2: ModifiedCNN architecture with BatchNorm and Global Average Pooling.

Layer	Output shape
Input	$3 \times 224 \times 224$
Conv(3,32) + BN + ReLU	$32 \times 224 \times 224$
MaxPool(2)	$32 \times 112 \times 112$
Conv(32,64) + BN + ReLU	$64 \times 112 \times 112$
MaxPool(2)	$64 \times 56 \times 56$
Conv(64,128) + BN + ReLU	$128 \times 56 \times 56$
MaxPool(2)	$128 \times 28 \times 28$
Global Average Pooling	128 features
FC(128 $\rightarrow$ 256) + ReLU + Dropout(0.5)	256
FC(256 $\rightarrow$ 2)	logits

This model contains only **0.4 million parameters**, more than 25x fewer than the baseline, yet maintaining similar representational depth. The use of BatchNorm and GAP typically leads to smoother optimization, reduced overfitting, and improved calibration.

3.5 Training Procedure

All models were trained using a standardized pipeline to ensure a fair comparison between architectures. Optimization was performed using the Adam algorithm [Kingma and Ba, 2014], a widely used variant of stochastic gradient descent that adapts the learning rate for each parameter based on estimates of first- and second-order moments of the gradients. This was used since Adam typically converges faster and more stably than vanilla SGD, which makes it well-suited for medical imaging tasks with noisy gradients. We used a learning rate of 1×10^{-4} , which provided stable training without requiring aggressive learning-rate tuning. The batch size was set to 32, balancing computational efficiency with sufficient gradient diversity on a single consumer-grade GPU.

A key challenge in training on SLICE-3D is the extreme class imbalance, with fewer than 1% of the lesions in the dataset being malignant [Johnson and Khoshgoftaar, 2019]. To address this, we employed a weighted cross-entropy loss that assigns higher importance to malignant samples. The weights were computed directly from the class frequencies in the training split. This ensures that the model does not simply learn to predict the majority (benign) class. To avoid overly large weight values destabilizing training, we used a “balanced” weighting mode that clips extreme weights while maintaining emphasis on the minority class.

In addition to training on the original class distribution, we also trained each model using an oversampling factor of 5 for the malignant class. These oversampled runs were included as part of the main experimental setup in order to directly compare performance with and without increased minority-class exposure.

Training was conducted for 10 epochs. To further stabilize training and prevent overfitting, we used a ReduceLROnPlateau scheduler that monitors the validation AUC and halves the learning rate if performance plateaus. This allows the model to refine its decision boundary during later stages of training.

All data augmentations were applied only during training to improve generalization (Section 3.2), while validation and test images were processed using deterministic resizing and normalization to ensure fair evaluation. A fixed random seed was used across experiments to guarantee reproducibility.

Overall, the training pipeline follows standard best practices for deep learning under severe class imbalance: adaptive optimization (Adam), careful class reweighting, moderate regularization through augmentation, and learning-rate scheduling based on validation performance.

3.6 Patient-Level “Ugly Duckling” Methods

In clinical practice, dermatologists often assess a patient’s lesions in context rather than individually. This principle, known as the *ugly duckling sign*, states that a lesion appearing noticeably different from a patient’s other lesions is more likely to be malignant [Scope et al. 2008]. The SLICE-3D dataset allows us to replicate this approach computationally, since each patient has multiple associated lesion crops. In this study, we explore two strategies for translating this clinical heuristic into model-based patient-level predictions.

3.6.1 Embedding-Based Outlier Detection

The first approach operates on the feature embeddings extracted from the penultimate layer of the network. For a given patient, we compute a feature vector for each lesion and then quantify how much each lesion deviates from the patient’s overall feature distribution. Specifically, we compute the distance between each lesion embedding and the patient’s mean embedding, treating large deviations as potential “ugly ducklings”. The lesion with the greatest feature-distance is then flagged as the most suspicious.

This method depends critically on the quality of the learned embeddings. If the network does not capture semantically meaningful differences between benign and malignant lesions, the resulting distances may reflect noise, augmentation artifacts, or irrelevant texture patterns rather than clinically meaningful outliers. As a result, embedding-based detection can deteriorate when using shallow or undertrained networks.

3.6.2 Probability-Based Ranking

The second approach uses the models predicted probability of malignancy for each lesion. For each patient, the lesion with the highest predicted probability is treated as the most suspicious. Conceptually, this mimics a triage workflow: from a set of lesions, highlight the one most likely to require clinical attention.

Compared to the embedding-based method, this strategy only considers the final classification output, which often contains stronger cues for malignancy than intermediate embeddings. While it does not explicitly model relative differences among a patient’s lesions, it offers a simple and robust approximation of patient-level screening.

Together, these two approaches provide complementary perspectives: one emphasizes intra-patient variation (embedding-based), while the other relies on absolute malignancy likelihood (probability-based).

4 Results

4.1 Lesion-Level Classification Performance

In Table 3 we summarize lesion-level classification performance for the Baseline CNN and the Modified CNN, with and without oversampling. Because of the strong class imbalance in the dataset, all models struggled to identify malignant lesions, resulting in precision, recall, and F1-scores near zero.

Overall the Baseline CNN achieved the weakest performance with an AUC of 0.5173. It frequently collapsed to predicting only the benign class, indicating that it was unable to learn stable or discriminative embeddings under such extreme imbalance. Nonetheless, it served its purpose as a functional reference model, confirming that the preprocessing and training pipeline operated correctly.

Among all architectures, the Modified CNN without oversampling achieved the highest AUC score of **0.6663**, but still failed to correctly identify a single malignant lesion (it found 0 positives), as shown in Figure 4. The modified architecture displayed more stable convergence and slightly improved feature aggregation, yet its limited depth and the low amount of malignant samples still led to a clear benign-class bias.

Table 3: Lesion-level classification metrics for the Baseline CNN and the Modified CNN models.

Model	AUC	Sens.@95%Spec.	Precision	Recall
Baseline	0.5173	0.0000	0.0000	0.0000
Baseline with Oversampling	0.5267	0.0000	0.0024	0.0380
Modified	0.6663	0.0000	0.0000	0.0000
Modified with Oversampling	0.6122	0.0000	0.0000	0.0000

The training curves in Figure 3 further illustrate these trends. The Modified CNN demonstrated smoother convergence and consistently higher validation AUC compared to the baseline, though neither architecture succeeded in recovering malignant lesions.

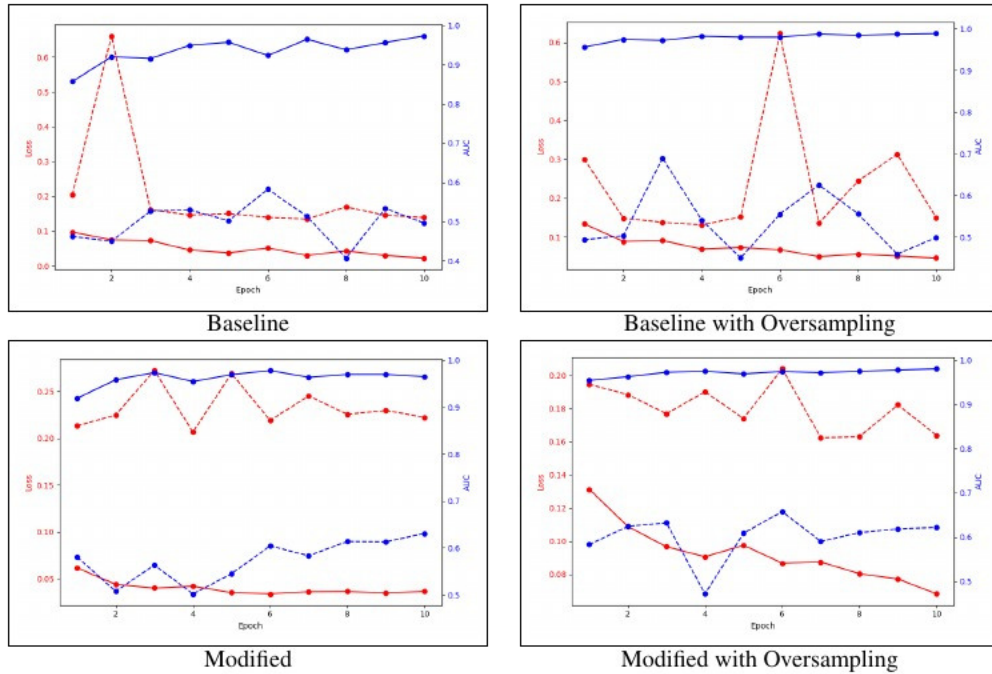


Figure 3: Training and validation loss/AUC for the Baseline CNN and the Modified CNN. Training is the full line, while the dotted line is the validation.

4.1.1 Effect of Oversampling

Oversampling the malignant class by a factor of 5 produced mixed results. For the Baseline CNN oversampling led to a much higher number of false positives and only a slight improvement in AUC. For the Modified CNN oversampling reduced the AUC, and did not recover any additional malignant lesions, still producing zero true positives. This is illustrated in the confusion matrices in Figure 4.

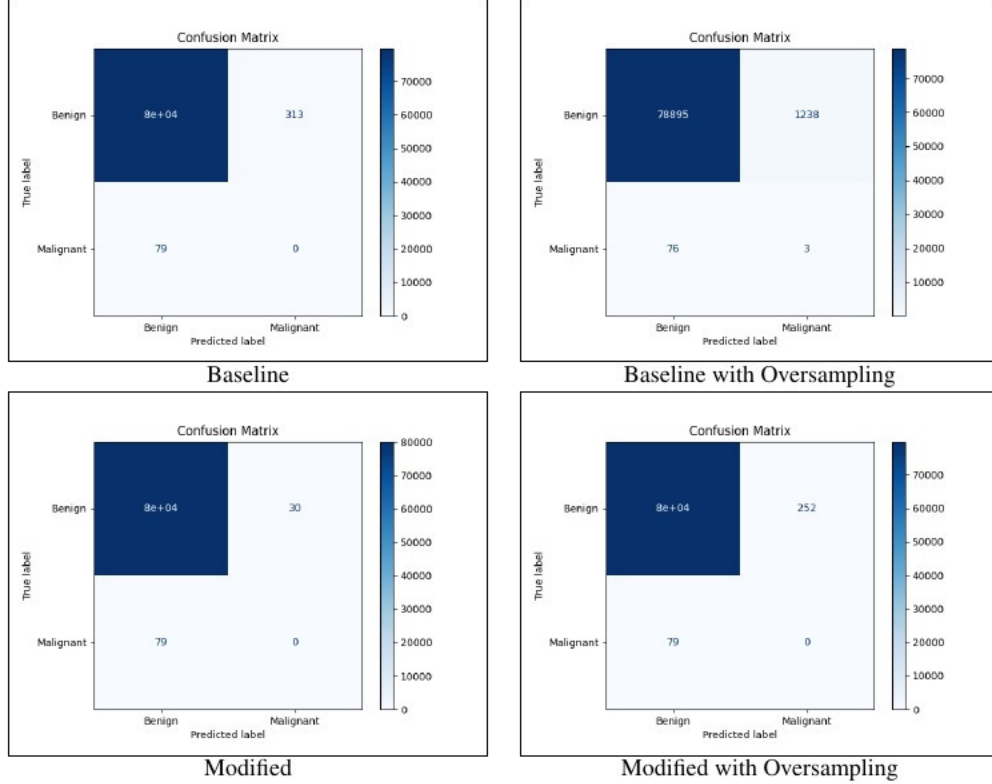


Figure 4: Confusion matrices for the Baseline CNN and the Modified CNN.

These results indicate that oversampling alone isn’t sufficient to solve this kind of extreme class imbalance and may even weaken training slightly.

4.2 Patient-Level “Ugly Duckling” Analysis

To evaluate whether the models could highlight the single most suspicious lesion per patient, we tested two approaches: (1) selecting the most outlying embedding, and (2) selecting the lesion with the highest malignant probability. Table 4 reports the number of correctly identified malignant lesions among the 1032 patients.

We only did this for the Modified CNN due to our Baseline CNN not being set up for embeddings. The Modified CNN had a extremely low patient-level recall, only correctly identifying 4 malignant lesions using the embedding-based strategy (0.39%), and just one lesion using probability ranking (0.10%). Oversampling did not change this outcome.

Table 4: Patient-level ugly-duckling results. Recall is the proportion of patients for whom the selected lesion was actually malignant.

Model	Embedding-based Recall	Probability-based Recall
Modified CNN	0.0039	0.0010
Modified CNN with Oversampling	0.0039	0.0010

5 Discussion

The results of this study highlight both the potential and the limitations of lightweight convolutional neural networks for melanoma classification on the SLICE-3D dataset. Across all experiments, models consistently learned to recognize benign lesions, but detection of malignant lesions remained extremely limited. Even with loss reweighting and oversampling, malignant recall remained near zero (Tables 3, 4), reflecting the severe class imbalance in the dataset.

Comparing the two architectures, the Modified CNN consistently achieved higher lesion-level AUC than the Baseline CNN (0.6663 vs 0.5173 without oversampling), demonstrating that modest architectural improvements such as batch normalization and global average pooling can stabilize training and slightly improve feature learning. However, sensitivity and patient-level metrics remained very low (Tables 3, 4), showing that the limited depth and lack of pretrained features prevent effective detection of rare malignant cases. Oversampling (factor 5) produced minor changes in AUC but did not improve recall or patient-level detection, suggesting that simply increasing the number of malignant examples is insufficient to overcome the underlying challenge.

The patient-level “ugly duckling” analysis offered additional insight into model behavior. Both embedding-based and probability-based methods identified very few malignant lesions, with the probability-based approach slightly outperforming the embedding-based one (Table 4). This indicates that the learned embeddings from shallow networks are insufficiently discriminative for intra-patient outlier detection, while final-layer predictions carry weak malignant signals that can be exploited for ranking lesions. True utility of the ugly-duckling heuristic would likely require deeper networks or pretrained feature extractors capable of generating semantically meaningful embeddings.

Several limitations of this work should be acknowledged. The models’ limited capacity constrained performance on this highly imbalanced and visually subtle dataset. Computational constraints prevented the use of larger or pretrained backbones, which are commonly adopted in ISIC challenge-winning pipelines [Codella et al., 2019]. Additionally, augmentation and input resolution were relatively modest, potentially restricting exposure to diverse patterns. Finally, while weighted cross-entropy and oversampling were implemented to mitigate class imbalance, more sophisticated strategies—such as focal loss or effective number reweighting—may improve sensitivity to malignant lesions.

Despite these limitations, the study demonstrates a complete deep-learning workflow, from data preparation through lesion- and patient-level evaluation. The results provide a clear baseline, highlighting where shallow CNNs fail and offering guidance for future work toward more capable architectures and imbalance-handling methods.

6 Conclusion

This study examined lesion-level and patient-level melanoma classification using two lightweight CNN architectures on the SLICE-3D dataset. Despite careful preprocessing, data balancing, and weighted training, malignant detection remained extremely limited, and patient-level ugly-duckling identification succeeded only in a handful of cases. These outcomes reinforce the understanding that shallow models struggle on datasets with extreme imbalance and subtle visual cues.

While the Modified CNN provided modest improvements over the Baseline CNN in terms of AUC, sensitivity and patient-level recall remained near zero, and oversampling had minimal effect. These findings indicate that more powerful approaches—such as deeper networks, pretrained feature extractors, or advanced imbalance-mitigation strategies—are necessary for clinically meaningful performance.

Importantly, the study validates the end-to-end pipeline for data curation, preprocessing, model training, and evaluation on a large real-world dermatology dataset. The results serve as a baseline for future experimentation and underscore the practical challenges of medical imaging with severe class imbalance, providing clear motivation for subsequent work leveraging higher-capacity models and richer features.

7 Future Work

Several directions could strengthen and extend this project further. First, using pretrained convolutional backbones such as MobileNetV2, EfficientNet-B0, ResNet-18 or ResNet-34 could substantially improve feature extraction while keeping computational demands manageable. These architectures typically converge faster and provide more robust representations compared to models trained from scratch.

Another avenue is to incorporate alternative loss functions designed for severe class imbalance. Although this work uses class-weighted cross-entropy, methods such as focal loss or effective number of samples weighting may help the model prioritize difficult malignant cases and stabilize learning under skewed distributions.

Finally, the “ugly-duckling” lesion comparison could be expanded using richer embedding spaces. Transformer-based models such as DINOv2 could provide more discriminative similarity metrics, enabling more informative pairwise or neighbourhood-based lesion analysis.

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